딜러와 플레이어의 대결

-블랙잭-

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02. 게임 구현과정

03. 코드 설명

04. 게임 실행

블랙잭, 어떻게 하지?

주요 용어:

1. ***Hit(히트)*** - 카드 1장 추가, ***Stay(스테이)***- 카드 더 받지 않음.
2. ***Burst(버스트)*** : 플레이어 또는 딜러의 카드 합이 21을 초과한 경우 => 패
3. ***push(푸시)*** : 플레이어와 딜러 카드 합이 같은 경우 => 무승부

기본 규칙:

1. 블랙잭은 각 플레이어(1명 이상)와 딜러의 경쟁
2. 각자 최종 카드들의 합이 ***21과 같거나 가장 가까운*** 사람이 승리 (초과(버스트)시, 패배)
   - 무승부(푸시)시, 배팅금액은 본인에게 그대로 돌아감.
3. 카드 계산: 숫자 2~9는 고유 숫자 그대로, 10/J/Q/K는 모두 값 10으로 통일
   - A= 1 or 11. BUt, 우리 프로그램에서는 11이 우선 적용된 경우의 합이 21을 초과할시, 1 적용으로 바뀜.
4. 처음에 카드를 2장씩 받으며, 딜러는 두 카드 중 하나만 플레이어들에게 공개
5. 이후, 플레이어가 히트 혹은 스테이를 선택하면
   딜러도 히트 혹은 스테이를 선택하되 플레이어는 그 여부를 알 수 없음.
6. 마지막으로, 딜러의 카드가 공개!! 바로, 승패여부가 결정됨~!!

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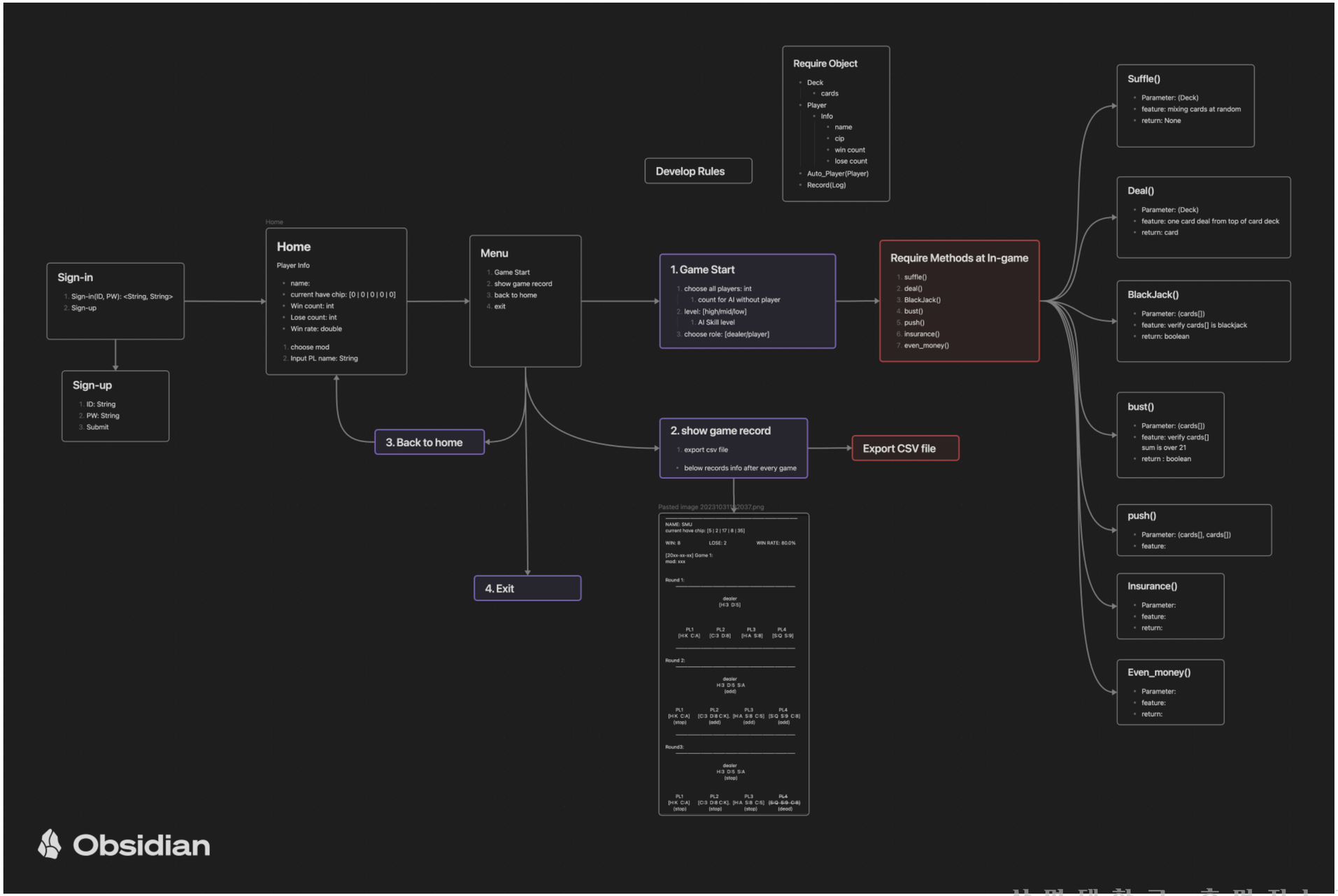
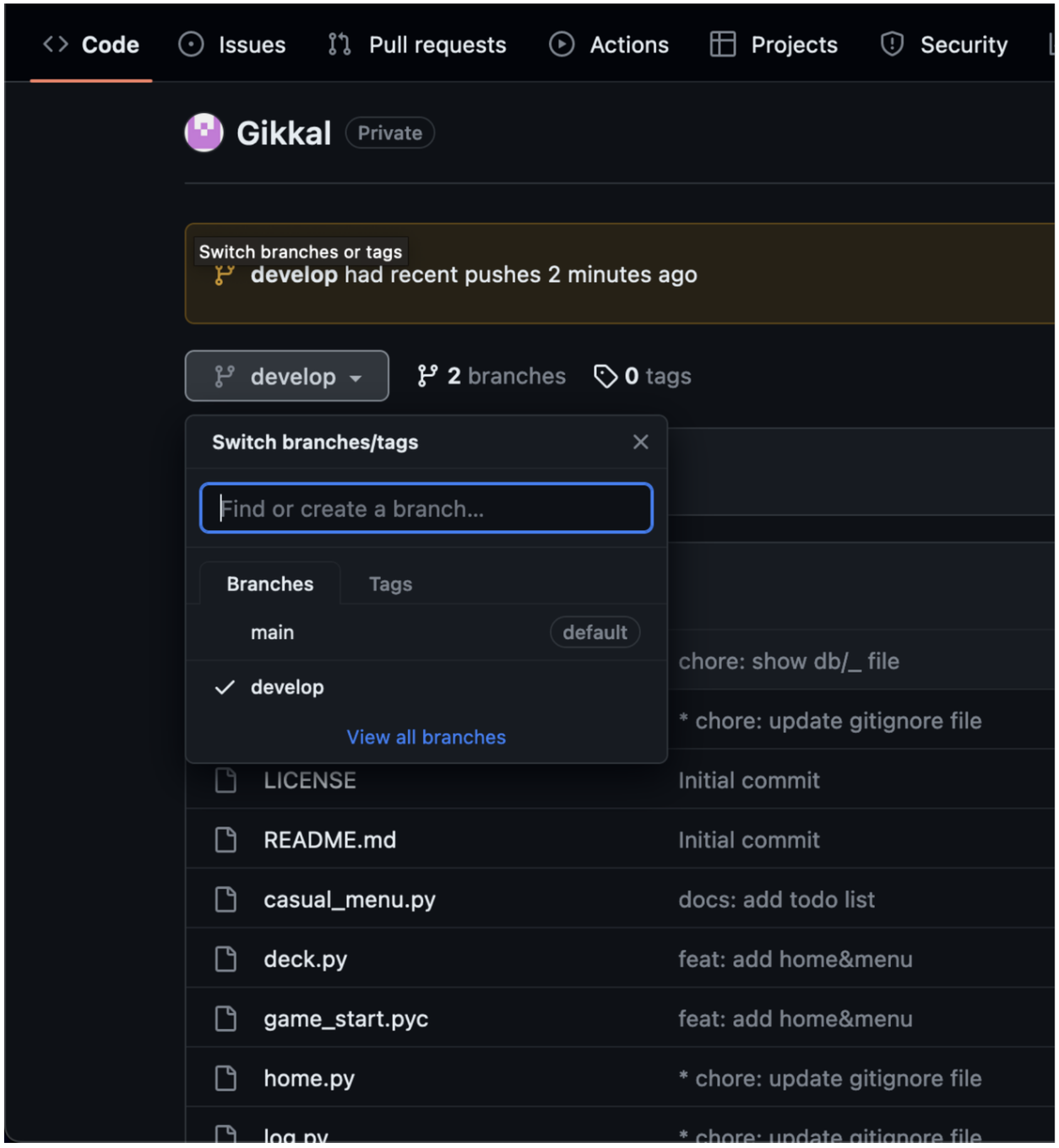


초반 회의 내용

옵시디언 이용

개발자 플랫폼

GitHub 사용



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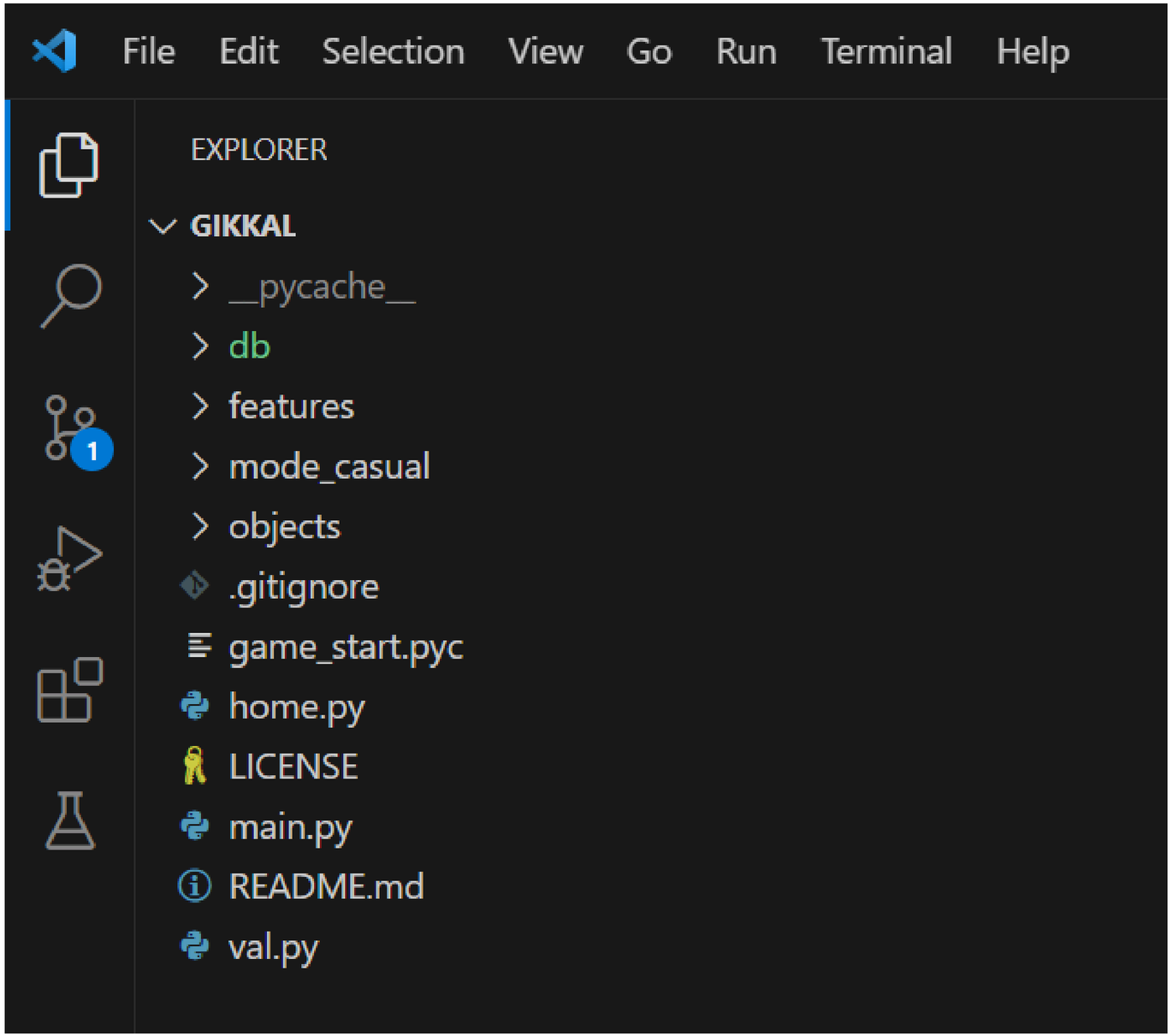

File Edit Selection View Go Run Terminal Help
EXPLORER
GIKKAL
__pycache__
db
features
mode_casual
objects
.gitignore
game_start.pyc
home.py
LICENSE
main.py
README.md
val.py

전체적인 구조

main.py > ...

```python
import home
import features.game as game
from objects.card import Card
from objects.player import Player

name = input("Please input your name : ")
pl = home.init_db(name)
home.lobby(pl)
```

부분 설명

main.py

- 모듈을 가져오고 게임이나 애플리케이션과 관련된 일부 코드 포함
- 플레이어 정보 설정, 데이터베이스 초기화, 애플리케이션 내 로비 탐색

실행 버튼을 누르면~! =>

```
Please input your name :
```

```
Please input your name : king
Sign-up: king

Player Info
-------------------------------------
name : king
current chip : [10 | 0 | 0 | 0 | 0 | 0]
win count : 0
lose count : 0
win rate : 0.0%
-------------------------------------



Choice game mode
-------------------------------------
1. Casual
2. !Basic! (Not yet...)
3. show money($)
-------------------------------------
```

home.py > init_db

```python
import os
import mode_casual.casual_menu as casual_menu
import features.logs as logs
import features.calc_chip as calc
from objects.player import Player
from val import CLI_I

def init_db(name):
    file_path = logs.get_db_file_path(name)
    if os.path.exists(file_path):
        print(f"Sign-in: {name}")
        pl = logs.player_txt_to_object(file_path)
        return pl

    print(f"Sign-up: {name}")
    f = open(file_path, "w")
    pl = Player(name, [10, 0, 0, 0, 0, 0])

    f.write(logs.player_object_to_txt(pl))
    f.close()
    return pl


def choose_mod(pl):
    mode = input(CLI_I.DB_MENU)
    if mode == "1":
        casual_menu.choice_menu(pl)
    elif mode == "2":
        print("Implementing...")
        choose_mod(pl)
    elif mode == "3":
        print('current money: ', (int)(calc.calc_money(pl.get_curr_chip())),'$')
        choose_mod(pl)

def lobby(pl):
    print(logs.player_object_to_txt(pl))
    choose_mod(pl)
```

```
Choice game mode
-------------------------------
1. Casual
2. !Basic! (Not yet...)
3. show money($)
-------------------------------
1

choice :
-------------------------------
1. Game Start
2. Show game record
3. Back to home
4. Explain rule
5. Exit
-------------------------------
```

mode_casual > casual_menu.py > ...

```python
import home
import features.logs as logs
from val import (CLI_I, FILE)
import objects.result as game

def choice_menu(pl):
    menu = input(CLI_I.CHOICE_MENU)
    if menu == "1":
        game.game_start(pl)
    elif menu == "2":
        num = input(CLI_I.EXPORT_FILE_MENU)
        if num == '1':
            export_file(FILE.CSV, pl.name)
        elif num == '2':
            export_file(FILE.ZIP, pl.name)
        elif num == "3":
            home.lobby(pl)
    elif menu == "3":
        logs.back_to_home(pl)
    elif menu == "4":
        print(CLI_I.EXPLAIN_RULE)
        choice_menu(pl)
    elif menu == "5":
        exit(0)

def export_file(category,  name):
    message = ""
    if category == FILE.CSV:
        message = logs.export_txt_to_csv(name, logs.get_db_file_path(name))
    elif category == FILE.ZIP:
        message = logs.download_all_log(name, logs.get_db_file_path(name))
    print(f"Successfully export {category} file!!")
    print(f"{category} file dir: ", message)
```

```
choice :
----------------------------------------
1. Game Start
2. Show game record
3. Back to home
4. Explain rule
5. Exit
----------------------------------------
1
```

```
----------------------------------------
1. Select Player Count and Game Start
2. Back to home
1
```

```
Player Count(1~3): 2
```

```
Input bet price: (1~1000): 500
```

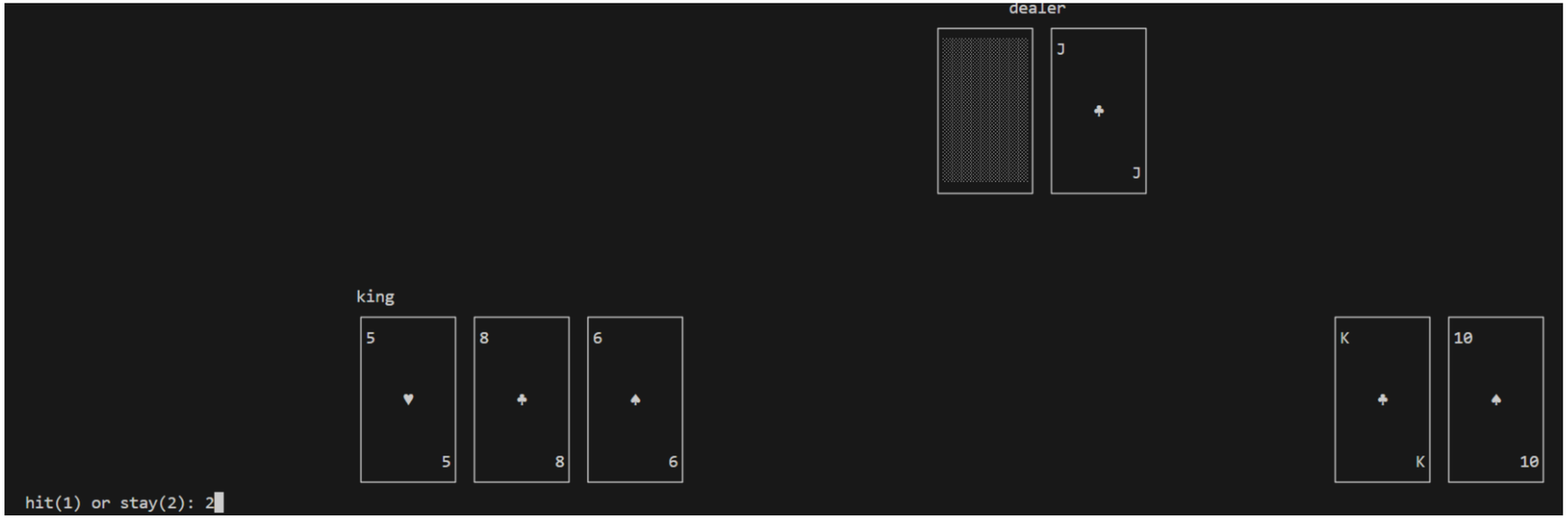
```
Input bet price: (1~1000): 500

                                                dealer
                                                J
                                                ♣
                                                J

king
5                8
♥                ♣
5                8

                                                K                10
                                                ♣                ♠
                                                K                10

hit(1) or stay(2):
```

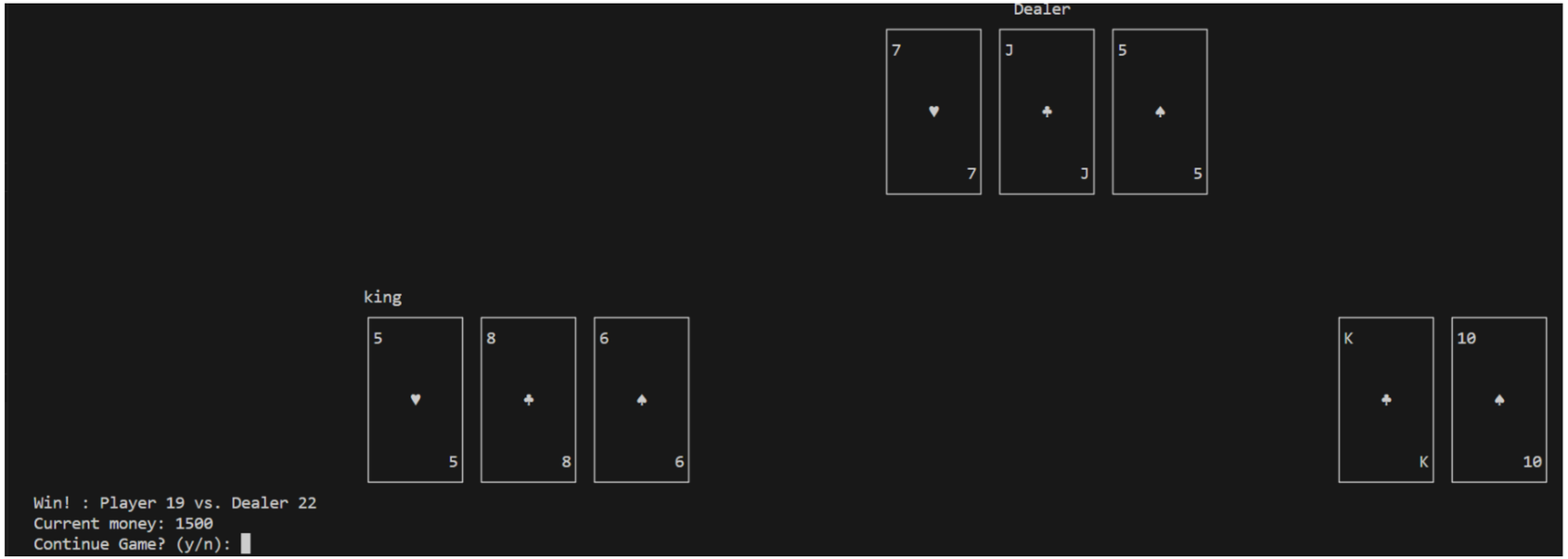
objects > card.py > Card

```python
class Card(object):

    card_values = {
        'Ace': 11,  # value of the ace is high until it needs to be low
        '2': 2,
        '3': 3,
        '4': 4,
        '5': 5,
        '6': 6,
        '7': 7,
        '8': 8,
        '9': 9,
        '10': 10,
        'Jack': 10,
        'Queen': 10,
        'King': 10
    }

    def __init__(self, suit, rank):
        """
        :param suit: The face of the card, e.g. Spade or Diamond
        :param rank: The value of the card, e.g 3 or King
        """
        self.suit = suit.capitalize()
        self.rank = rank
        self.points = self.card_values[rank]
        self.name = f'{rank} of {suit}'
```

```python
def ascii_version_of_card(*cards, return_string=True):
    """
    Instead of a boring text version of the card we render an ASCII
    :param cards: One or more card objects
    :param return_string: By default we return the string version o
    keep it as a list so that the dealer can add a hidden card in f
    """
    # we will use this to prints the appropriate icons for each car
    suits_name = ['Spades', 'Diamonds', 'Hearts', 'Clubs']
    suits_symbols = ['♠', '♦', '♥', '♣']

    # create an empty list of list, each sublist is a line
    lines = [[] for i in range(9)]

    for index, card in enumerate(cards):
        # "King" should be "K" and "10" should still be "10"
        if card.rank == '10':  # ten is the only one who's rank is
            rank = card.rank
            space = ''  # if we write "10" on the card that line wi
        else:
            rank = card.rank[0]  # some have a rank of 'King' this
            space = ' '  # no "10", we use a blank space to will th
        # get the cards suit in two steps
        suit = suits_name.index(card.suit)
        suit = suits_symbols[suit]

        # add the individual card on a line by line basis
        lines[0].append('┌─────────┐')
        lines[1].append('│{}{}      │'.format(rank, space))  # use
        lines[2].append('│         │')
        lines[3].append('│         │')
        lines[4].append('│    {}   │'.format(suit))
        lines[5].append('│         │')
        lines[6].append('│         │')
        lines[7].append('│      {}{}│'.format(space, rank))
        lines[8].append('└─────────┘')
```

```python
def ascii_version_of_hidden_card(*cards):

    """
    Essentially the dealers method of print ascii cards. This method hides the first card, shows it flipped over
    :param cards: A list of card objects, the first will be hidden
    :return: A string, the nice ascii version of cards
    """
    # a flipper over card. # This is a list of lists instead of a list of string becuase appending to a list is better then adding a string
    print('\t'*13,  end='')
    print('dealer')
    lines = [['\t'*12 + '┌─────────┐'], ['\t'*12 + '│▒▒▒▒▒▒▒▒▒│'], ['\t'*12 + '│▒▒▒▒▒▒▒▒▒│'], ['\t'*12 + '│▒▒▒▒▒▒▒▒▒│'], ['\t'*12 + '│▒▒▒▒▒▒▒▒▒│'],
             ['\t'*12 + '│▒▒▒▒▒▒▒▒▒│'], ['\t'*12 + '│▒▒▒▒▒▒▒▒▒│'], ['\t'*12 + '│▒▒▒▒▒▒▒▒▒│'], ['\t'*12 + '└─────────┘']]
```

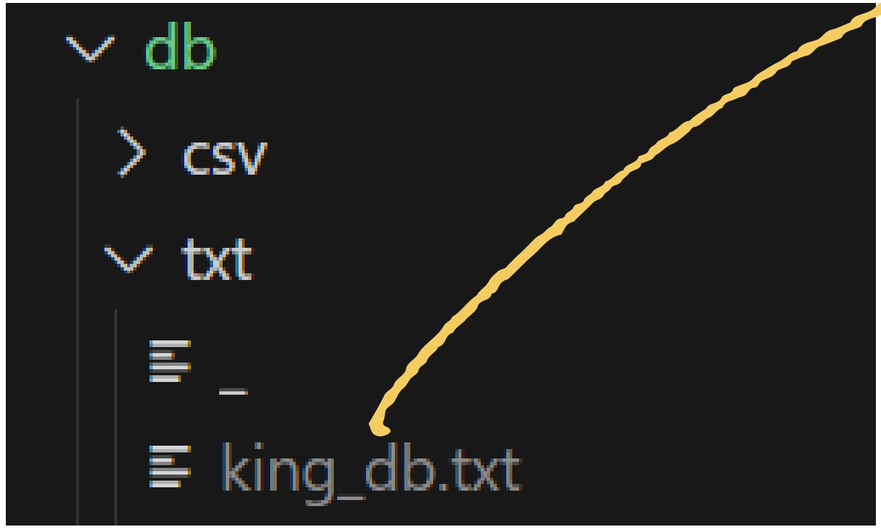




- 딜러의 히트 또는 스테이 여부는 알 수 없음!
- 마지막에는 모든 카드가 공개!!
- 딜러의 카드 버스트 되었으므로 플레이어 승리!
- 현재 보유 머니: 1500

```
Continue Game? (y/n): n
Bye.
```

게임 종료 후,
다시 로그인을 하면?



저장된 플레이어의 이전 기록을 볼 수 있다!!!!

```
Please input your name : king
Sign-in: king

Player Info
---------------------------------------
name : king
current chip : [15 | 0 | 0 | 0 | 0 | 0]
win count : 1
lose count : 0
win rate : 100.0%
---------------------------------------


Choice game mode
----------------------------------------
1. Casual
2. !Basic! (Not yet...)
3. show money($)
----------------------------------------
```

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File Edit Selection View Go Run Terminal Help

Gikkal

EXPLORER

GIKKAL

- zip
- _
- s_db.txt U
- features
  - __pycache__
  - calc_chip.py
  - game.py
  - logs.py
- mode_casual
  - __pycache__
  - casual_menu.py
- objects
  - __pycache__
  - action.py
  - betting.py
  - card.py
  - deck.py
  - player.py
  - playgame.py
  - result.py
- .gitignore
- game_start.pyc
- home.py
- LICENSE
- main.py
- README.md
- val.py

OUTLINE

TIMELINE

VS CODE PETS

Welcome | main.py

main.py > ...

```python
import home
import features.game as game
from objects.card import Card
from objects.player import Player

name = input("Please input your name : ")
pl = home.init_db(name)
home.lobby(pl)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, **/*.ts, !**/node_modules/**)

No problems have been detected in the workspace.

바탕 화면 보기

develop* 0 0 0

Ln 8, Col 15 Spaces: 4 UTF-8 CRLF Python 3.11.2 64-bit

감사합니다

기깔나는팀

팀장 이선역　　팀원 강동현

팀원 송예린　　팀원 유세리